
CSE6364- Lecture 8
Software Change Management 1

Learning Outcomes

After completing this lecture, you should be able to:

- Describe the different changes that could be done on a software and their relationships
- Describe the management of software changes
- Explain the change control process, and the documentation of the changes

Lecture 8 - Outline

- Drivers for Change
- Software Change Management
- Change Control Processes & Documentation

Drivers for Change

Quote

The tendency to change is necessary to optimally adjust to environmental requirements

[Damste, 74]

Sources of Changes

- New customer needs
- New market conditions
- Business reorganization
- Budgetary or scheduling constraints
- Performance or reliability issues
- New equipment
- Errors that must be repaired
- ...

Types of Changes

- Adaptive change
 - A change made in order to become suited to different conditions.
- Corrective change
 - A change made in order to remove faults.
- Perfective change
 - A change made in order to improve
- Preventive change
 - A change made in order to forestall or reverse deterioration.

Corrective Change

- Refers to modification initiated by defects in the software.
- Defects can result from
 - **Design errors** occur when, for e.g., changes made to the software are incorrect, incomplete, wrongly communicated or the change request is misunderstood.
 - **Logic errors** result from invalid tests and conclusion, incorrect implementation of design specifications, faulty logic flow or incomplete testing of data.
 - **Coding errors** are caused by incorrect implementation of detailed logic design and incorrect use of the source code logic. Defects are also caused by data processing errors and system performance errors.

Adaptive Change

- A change driven by the need to accommodate modifications in the environment of the software system.
- The term environment in this context refers to the totality of all conditions and influences which act from outside upon the system, for e.g. business rules, government policies, work patterns, software and hardware operating platforms.
- A change to the whole or part of this environment will warrant a corresponding modification of the software.

Perfective Change

- Changes undertaken to expand the existing requirements of a system.
- A successful piece of software tends to be subjected to a succession of changes resulting in an increase in its requirements.
- Expansion in requirements can take the form of enhancement of existing system functionality or improvement in computational efficiency. For e.g. providing a MIS with a data entry module or a new message handling facility.

Preventive Change

- The long term effect of corrective, adaptive and perfective changes is expressed in Lehman's law of increasing entropy.
- Work done on a software system to address problems of deteriorating structure is known as preventive change.
- To prevent malfunctions or to improve maintainability of the software.
- E.g. code restructuring, code optimisation and documentation updating.

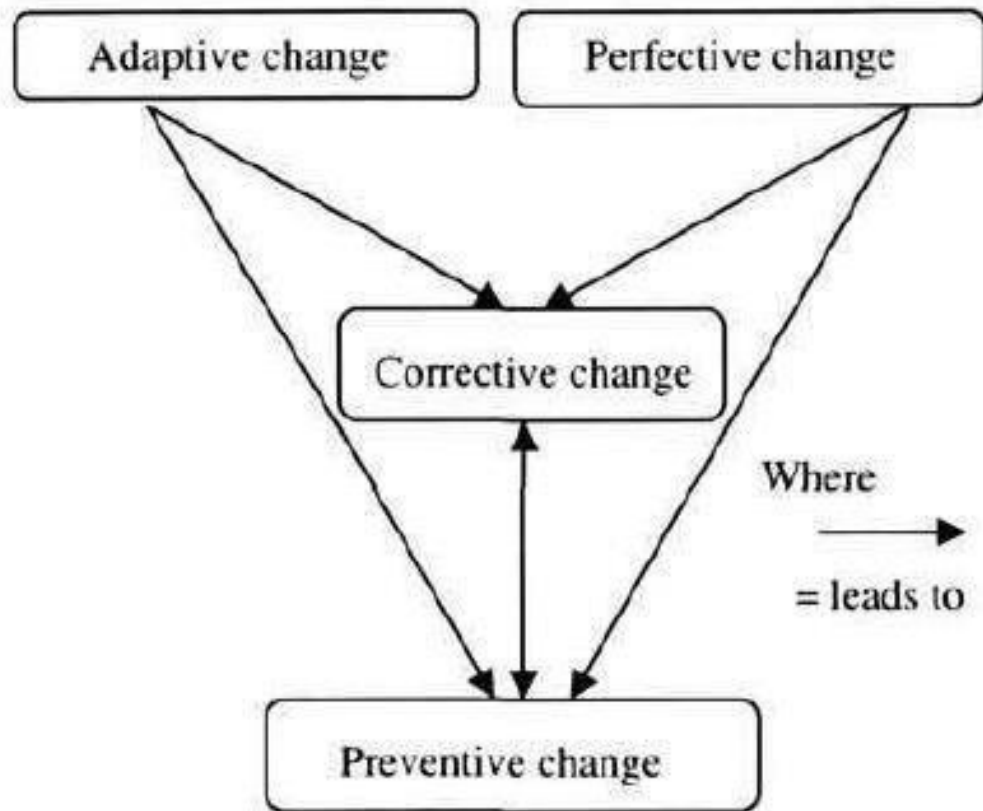
Preventive Change

- Unforeseen ripple effects imply that a change to one part of a program may affect other sections in an unpredictable fashion, thereby leading to a distortion in the logic of the system.
- This is due to the lack of time to carry out a thorough impact analysis before effecting a change.
- Program slicing techniques and concepts such as modularization and information hiding can be used to address these problems.

Importance of Categorizing S/w Changes

- In principle, software maintenance activities can be classified individually. In practice, they are usually intertwined.
- E.g. In the course of modifying a program due to the introduction of a new operating system (adaptive change), obscure bugs may be introduced. The bugs have to be traced and dealt with (corrective maintenance). Similarly, the introduction of a more efficient sorting algorithm into a data processing package (perfective maintenance) may require that the existing program code be restructured (preventive maintenance).

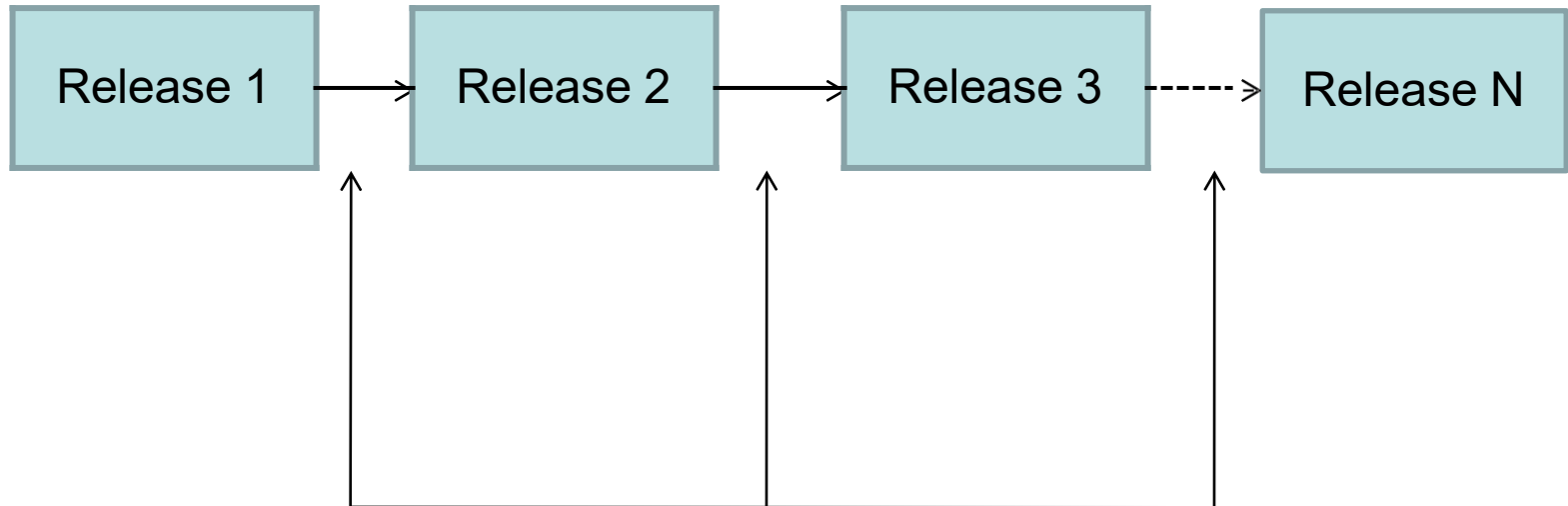
Potential Relation Between S/w Changes



Incremental Release

- The changes made to a software product are not always done all together. The changes take place incrementally, with minor changes usually implemented while a system is in operation.
- With customised software, change can often be effected as the need for it arises.
- For off-the-shelf packages, users normally have to wait for the next upgrade.

Incremental Product Release



Opportunity to introduce change

Software Change Management

Software Change Management

- Software systems are subject to continual change requests:
 - From users;
 - From developers;
 - From market forces.
- Change management is concerned with keeping track of these changes and ensuring that they are implemented in the most cost-effective way.

Change Management

Change Management is the discipline that guides how we:

- Prepare
- Equip
- Support

individuals to successfully adopt changes

Change Management Requirements

- Some means for users and developers to suggest required system changes
- A process to decide if changes should be included in a system
- Software to keep track of suggested changes and their status
- Software support for managing changing system configurations and building new systems

Change Management Processes

- A company needs to define a change management process and policy within the company's business structure and the software maintenance process.
- Change management is not an isolated process
 - the team must be clear on what, when, how, and why to carry it out.

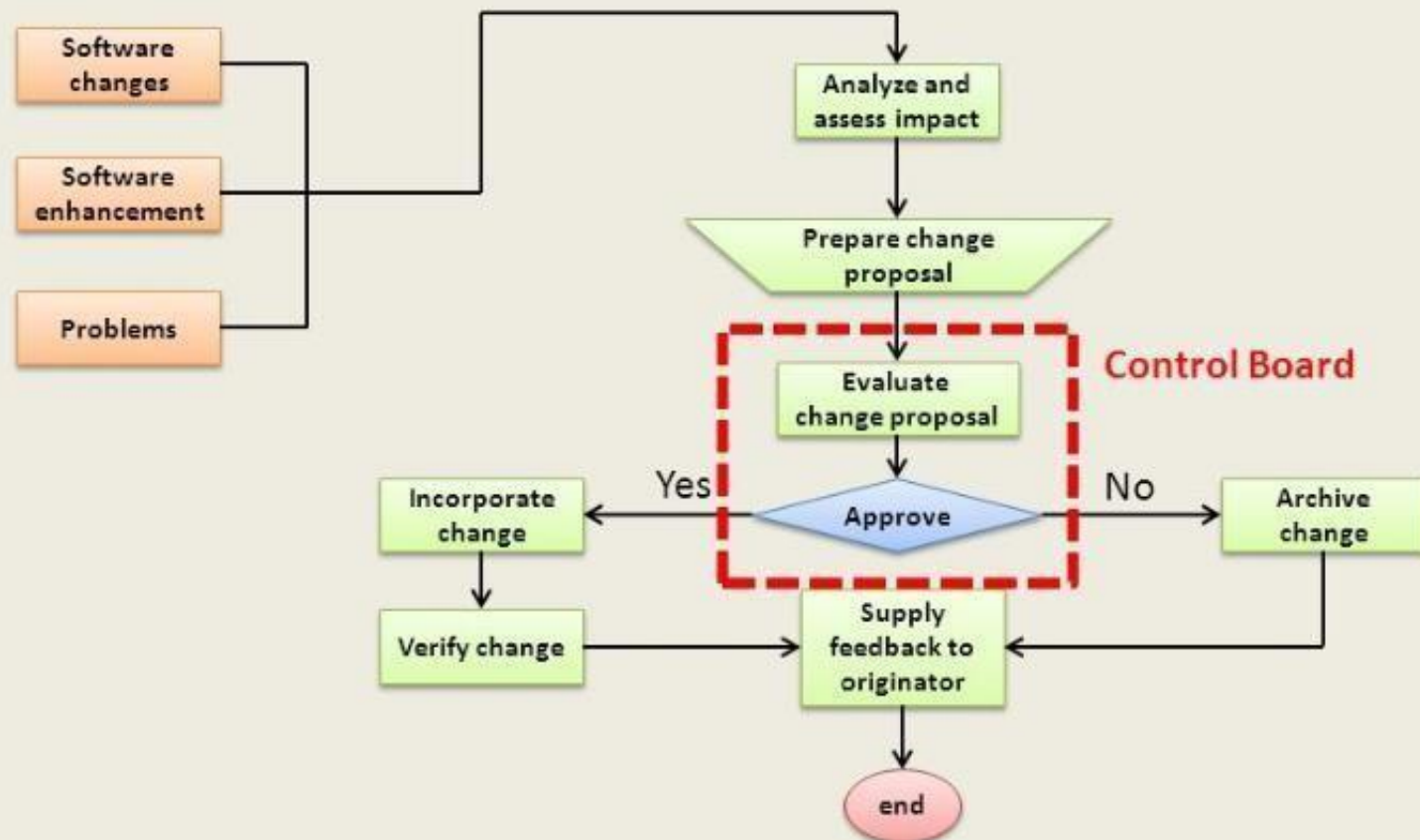
How to Achieve Successful Changes

- By assuring that all proposed changes are evaluated
- By prioritizing changes
- By requiring that all changes are thoroughly tested
- By having a back-out plan
- By ensuring that configuration management system is updated to reflect the effect of any changes

Change Control Processes & Documentation

Change Control Process

- A generic change control process [Berlack 1992]



Change Requests

- Change requests are requests for system changes from users, customers or management
- In principle, all change requests should be carefully analysed as part of the maintenance process and then implemented
- In practice, some change requests must be implemented urgently
 - Fault repair
 - Changes to the system's environment
 - Urgently required business changes

Creating a Request for Change

- Incidents that necessitate the change
- Description of how the change would be implemented
- The impact that the change would have on all associated systems
- A risk assessment
- Contact information for everyone involved in the change
- An outline of who will need to approve the request
- A backup plan to follow in case the change is not successful

Change Request Example

Change Request Form Example			
Project Name	House build		
Requested By	Project Sponsor	Date	18th April
Request No	5	Name of Request	Add Pool
Change Description	The addition of a 16x8 metre pool to the rear of the property. To include a fence around the pool to make it safe for children in the garden, and also a singular diving board.		
Change Reason	This is a new requirement from the project sponsor.		
Impact of change	Minimal groundwork required, but it is estimated the pool will cost \$25,000 and the fence a further \$2,500. Additionally, the completion of the project will be delayed by 2 weeks to complete this work.		
Proposed Action	Accept the change and the cost associated with it impact. However, complete this work after the rest of the house is finished. This way this work can be done after the owner has moved in and is not on the critical path.		
Status	In Review	Approved	Rejected
	YES		
Approval Date	N/A		
Approved By	N/A		

Change Request Example

Change Request Template:

We have used the following form template for change request:

Change Request		
Project:		Date:
Change Requestor:		Change No:
Change Category (Check all that apply):		
☐ Schedule	☐ Cost	☐ Scope
☐ Testing/Quality	☐ Resources	☐ Requirements/Deliverables
Does this Change Affect (Check all that apply):		
☐ Corrective Action	☐ Preventative Action	☐ Defect Repair
☐ Other	☐ Updates	
Describe the Change Being Requested:		
Describe the Reason for the Change:		
Describe all Alternatives Considered:		
Describe any Technical Changes Required to Implement this Change:		
Describe Risks to be Considered for this Change:		
Estimate Resources and Costs Needed to Implement this Change:		
Describe the Implications to Quality:		
Disposition:		
☐ Approve	☐ Reject	☐ Defer
Justification of Approval, Rejection, or Deferral:		
Change Board Approval:		
Name	Signature	Date

Impact Analysis

- Provides accurate understanding of the implications of a proposed change
- Helps the team make informed business decisions about which proposals to approve
- Three aspects:
 - Understand the possible implications of making the change
 - Identify all files, models, and documents
 - Identify the tasks and estimation of efforts

Checklist – Possible Implications

- ☐ Do any existing requirements in the baseline conflict with the proposed change?
- ☐ Do any other pending requirements changes conflict with the proposed change?
- ☐ What are the business or technical consequences of not making the change?
- ☐ What are the possible adverse side effects or other risks of making the proposed change?
- ☐ Will the proposed change adversely affect performance requirements or other quality attributes?

Checklist – Possible Elements Affected

- ☐ Identify any user interfaces changes, additions, or deletions required.
- ☐ Identify any changes, additions, or deletions required in reports, databases, or files.
- ☐ Identify the design components that must be created, modified, or deleted.
- ☐ Identify the source code files that must be created, modified or deleted
- ☐ Identify any changes required in build files or procedures.
- ☐ Identify existing unit, integration, system, and acceptance test cases that must be modified or deleted.

Estimating Effort

Effort
(Labor
Hours)

Task

- _____ Update the SRS or requirements database.
- _____ Develop and evaluate a prototype.
- _____ Create new design components.
- _____ Modify existing design components.
- _____ Develop new user interface components.
- _____ Modify existing user interface components.
- _____ Develop new user documentation and help screens.
- _____ Modify existing user documentation and help screens.
- _____ Develop new source code.
- _____ Modify existing source code.
- _____ Purchase and integrate third-party software.
- _____ Modify build files.
- _____ Develop new unit and integration tests.
- _____ Perform unit and integration testing after implementation.
- _____ Write new system and acceptance test cases.

Impact Analysis

- Identify the sequence in which the tasks must be performed
- Evaluate the change's priority by estimating the relative benefit, penalty, cost, and technical risk compared to other discretionary requirements

Change Proposal

- Results from Impact Analysis is compiled into a document called Change Proposal
- May contain additional details such as:
 - Possible resources that are needed to complete the change
 - Proposed timeline for implementation

Evaluation & Review

- Evaluate the request based on its priority, impact analysis
- Check whether requests related problems that have already been addressed
- Determine who would be responsible for fulfilling the request
- Assess implementer's ability to dedicate time to making the change

Evaluation Aspects

- Priority
- Budget
- Schedule
- Stability of operations
- Disruption to productivity
- Service Level Agreements

Change Control Board

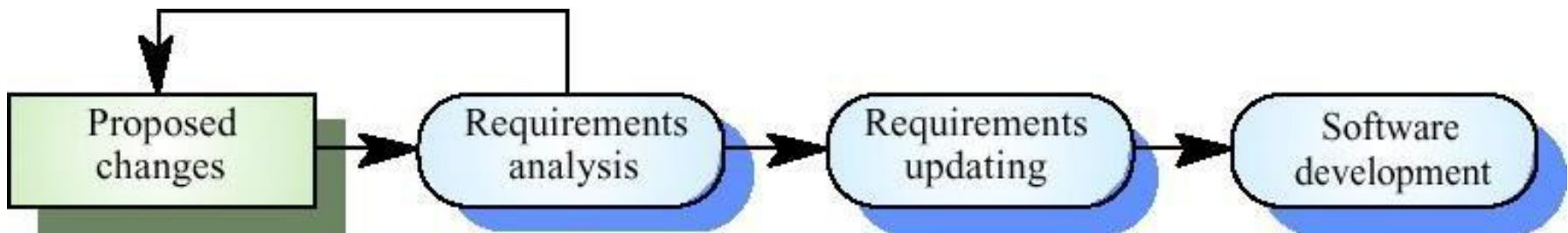
- A Change Control Board (CCB) is a committee that makes decisions regarding whether or not proposed changes to a software should be implemented.
- The members may include representatives from major stakeholders, project, and operations teams.
- Typically the board would include Subject Matter Experts (SME) and Technical Chiefs who can assess key technical and operational aspects that would be impacted by the decision.

Decision

- If the change proposal is not approved, the proposal is archived and the requesting parties are notified of the decision.
- If the change proposal is approved, the changes would be assigned to a maintenance team to implement the changes.
- It is possible for the decision to be deferred to a later date if the changes are not urgent, or the more details are required (and the proposal to be updated)

Change Implementation

- The approved changes would be implemented through a formal process of updating the requirements, followed by subsequent software engineering activities



Review and Testing

- The changes would require updating the documentation of the software
- The updated documentation must be reviewed to verify the correctness of documentation before updating the programs
- The program changes would be tested and confirmed before releasing the updated software to the customer.
- If the changes are not successful, the backup plan would be initiated.

End of lecture